Pulse — 16 March 2026

Headlines

- **Google's `longfellow-zk` open-sources ZKP age assurance** — Apache 2.0 licensed, already integrated into Google Wallet for apps like Bumble. Proves "over 18" without revealing identity data. Direct overlap with Signet's Bulletproofs age-range proof layer. `[COMPETITIVE-THREAT]`
- **EU age verification technical spec piloting in 5 countries** — Denmark, France, Greece, Italy, Spain running interim age verification ahead of EUDI Wallet mandate (end of 2026). Official GitHub repo with verifier UI. `[REGULATORY-SIGNAL]`
- **DIF ships `didwebvh-rs` release, ramps trusted-ai-agents working group** — Decentralized Identity Foundation active on identity assertions (CAWG), universal resolver, and a new "legally binding proof of personhood for verifiable credentials" lab. `[TECH-TO-WATCH]`
- **pablof7z building AI agent SDK infrastructure at speed** — 40+ pushes in 48 hours across `ai-sdk-context-management`, `ai-sdk-message-sanitizer`, `ai-sdk-system-reminders`, and `tenex-chat/tenex`. PR opened on `vercel/ai`. Nostr + AI convergence accelerating. `[SIGNET-OPPORTUNITY]`
- **NIP-XX AI Agent Messages (#2226) and NIP-AC DVM Agent Coordination (#2253) both active** — AI agent identity on Nostr is an open design space. Agent identity verification could become a Signet use case. `[SIGNET-OPPORTUNITY]`
- **Animo shipping EUDI wallet (paradym-wallet) aggressively** — Heavy PR and push activity on their wallet, EUDI docs, and OpenID4VC playground. First-mover on EU Digital Identity Wallet compliance. `[COMPETITIVE-THREAT]`
- **fiatjaf's `pyramid` — social proof of personhood** — "A wondrous furnace of communityzenship backed by a dynamic ladder of socialhood." Active development, 50 stars. Social-graph-based identity scoring on Nostr. `[TECH-TO-WATCH]`
- **NIP-XX: On-Chain EVM Identity Linking (#2262)** — Proposal to bridge Nostr keys to EVM addresses. Identity bridging is Signet's Kind 30476. `[SIGNET-OPPORTUNITY]`

Releases & Launches

Item	Details
notedeck release	jb55 published a new notedeck release (2026-03-15). The Nostr desktop browser continues rapid iteration.
`didwebvh-rs`	DIF released a new version of their Rust `did:webvh` implementation (2026-03-13). `[TECH-TO-WATCH]`
NIP-19 `limit` merged	PR #2264 by staab merged — adds `limit` to NIP-19. Minor but shows active spec maintenance.
pablof7z `context-management-webapp`	Made public (2026-03-16) — interactive showcase for AI SDK context management. New repos: `ai-sdk-message-sanitizer`, `ai-sdk-fs-tools`.

Protocol Movement

Proposal	Status	Relevance
NIP-XX: AI Agent Messages (#2226)	Open, active discussion (updated 2026-03-15)	Defines how AI agents communicate on Nostr. Agent identity verification is an untapped Signet use case. `[SIGNET-OPPORTUNITY]`
NIP-AC: DVM Agent Coordination v2 (#2253)	Open, active (updated 2026-03-15)	Data vending machine agent coordination — agents need identity/trust. `[SIGNET-OPPORTUNITY]`
NIP-XX: On-Chain EVM Identity Linking (#2262)	Open (2026-03-11)	Bridges Nostr keys to EVM addresses — overlaps with Signet Kind 30476 identity bridges. `[SIGNET-OPPORTUNITY]`
NIP-XX: Dual Encryption and ECDH Delegated Access (#2258)	Open (2026-03-07)	Delegated access patterns relevant to Signet's guardian delegation (Kind 30477). `[TECH-TO-WATCH]`
NIP-24 `shape` field (#2268)	Open (2026-03-16, alexgleason)	Kind 0 metadata extension — minor.
NIP-43 membership state (#2267)	Open (2026-03-15)	Group membership consistency — tangentially relevant to ring signature group management.

Proposal	Status	Relevance
W3C VC Data Model v2.1	New charter targeting July 2026	Render Method and Confidence Method heading to Recommendation. Signet credentials could align. `[TECH-TO-WATCH]`
Nostr DID Method	Active spec at nostrcg.github.io	W3C DID standard over Nostr keypairs — bridge between Signet's Nostr-native credentials and the broader VC ecosystem. `[SIGNET-OPPORTUNITY]`

Builder Activity

pablof7z — AI Agent Infrastructure Sprint

The most active builder in the window by far. Building a complete AI SDK middleware stack:

- `ai-sdk-context-management` — intelligent context window management for AI SDK v6
- `ai-sdk-message-sanitizer` — malformed message array cleanup
- `ai-sdk-system-reminders` — per-request reminder injection
- `ai-sdk-fs-tools` — portable filesystem tools with AGENTS.md support
- `tenex-chat/tenex` — appears to be an AI chat platform (heavy push activity)
- **PR opened on `vercel/ai`** — contributing upstream to the AI SDK

This is the clearest signal of Nostr + AI convergence from a key ecosystem builder. `[SIGNET-OPPORTUNITY]`

fiatjaf — pyramid + NIP stewardship

- `pyramid` seeing daily pushes — social proof system with "ladder of socialhood" concept (50 stars)
- Active NIP comment reviews across multiple proposals
- `nak` CLI tool updates (349 stars)
- Contributing to `fevela` (Nostr social client by dtonon)
- Active in `nostr-tools` discussions

Pyramid's social-proof identity model is philosophically adjacent to Signet's web-of-trust (Tier 2). `[TECH-TO-WATCH]`

jb55 — notedeck + hardware keys

- Notedeck release published 2026-03-15 (multiple PRs merged, issues closed)
- Starred `privkeyio/keep` and `keep-esp32` — hardware key storage project. Signal of interest in key management hardware. `[TECH-TO-WATCH]`
- `claude-agent-sdk-rs` repo exists — Rust Claude agent SDK wrapper

papiche — UPassport + Nostr identity

- 10+ pushes to `UPassport` (identity terminal API using Astroport, IPFS, strfry)
- `NIP-101` — decentralised trust system for Nostr with self-sovereign identity and peer-validated credentials
- Forked `rnostr` (Rust relay)
- Active on `Astroport.ONE` (Web3 platform with identity management)

papiche is building a parallel Nostr identity stack. NIP-101's peer-validated credentials directly overlap with Signet's vouch system. `[COMPETITIVE-THREAT]`

DIF (decentralized-identity org)

- `cawg-identity-assertion` — multiple issue comments, member additions, closures (2026-03-16)
- `universal-resolver` — continuous pushes (daily)
- `veramo` — DID framework updates
- `trusted-ai-agents` (13 stars) and `trusted-ai-agents-wg` (4 stars) — working group on AI agent identity
- `labs-legally-binding-proof-of-personhood-for-verifiable-credentials` — PoP lab

Animo

- `paradym-wallet` — heavy PR and push activity (daily since 2026-03-11)
- `paradym-eudi-docs` — EUDI integration documentation
- `openid4vc-playground` — OpenID for Verifiable Credentials playground
- `expo-digital-credentials-api` — Expo module for digital credentials (4 stars)

Media & Discourse

Moonshots EP #236 — Andrew Yang: Deepfakes Weaponising Elections

- Guest: Andrew Yang
- Date: 7 March 2026
- Key points: AI-generated deepfakes undermining democratic trust; UBI as response to AI job losses; open voting reform
- **Relevance:** Public discourse validating Signet's core threat model — deepfake-proof identity verification is not a niche concern, it's entering mainstream political conversation. `[REGULATORY-SIGNAL]`

Stephan Livera SLP723 — Bitcoin Payment Address Verification

- Guest: Keith Gardner (Branta)
- Date: ~19 February 2026
- Key points: Payment address phishing, verification solutions
- **Relevance:** Tangential — address verification is a different problem domain, but shows growing demand for cryptographic verification in payments.

Moonshots & Signals

Google `longfellow-zk` — Big Tech enters ZKP age verification

Google open-sourced its zero-knowledge proof libraries specifically for age assurance. Already integrated into Google Wallet. Apache 2.0 licence means anyone can build on it. This is the most significant competitive signal in the window — a FAANG company validating the exact problem space Signet occupies, but with a centralised wallet distribution channel. `[COMPETITIVE-THREAT]`

EU Age Verification Pilot (5 countries)

The EU Commission has an official GitHub repo (`eu-digital-identity-wallet/av-doc-technical-specification`) with technical specs and a verifier UI for interim age verification. Denmark, France, Greece, Italy, and Spain are piloting. This creates a regulatory forcing function — platforms operating in the EU will need age verification infrastructure by end of 2026. `[REGULATORY-SIGNAL]`

BetterAgeVerify — ML-based child safety

New project (January 2026) using on-device AI facial analysis for age verification. Claims 96% accuracy, GDPR/COPPA/BIPA compliance, immediate biometric deletion. Different approach from Signet (ML vs cryptographic proofs) but targeting the same child safety use case. `[COMPETITIVE-THREAT]`

DIF Proof of Personhood Lab

`labs-legally-binding-proof-of-personhood-for-verifiable-credentials` — DIF exploring legally binding PoP. This could set standards that Signet's Tier 3/4 professional verification aligns with or needs to comply with. `[TECH-TO-WATCH]`

AstroxNetwork Proof-of-Personhood

Decentralised AI-based PoP using face gestures and voice, fully on-device. No data leaves the device. Sybil-proof identity without central authority — similar philosophy to Signet but using biometrics rather than professional attestation. `[TECH-TO-WATCH]`

What This Means For Us

The ZKP age verification space is heating up fast. Google's `longfellow-zk` is the clearest validation that Signet's problem space matters, but it's also the strongest competitive signal. Google has distribution (Wallet integration) and brand trust. Signet's differentiator is decentralisation — no Google account required, no central authority, professional bodies as trust anchors. This positioning needs to be crystal clear in documentation and developer outreach.

AI agent identity on Nostr is an emerging design space with no winner yet. NIP-XX AI Agent Messages, NIP-AC DVM Agent Coordination, and pablof7z's AI SDK infrastructure sprint all point to the same conclusion: AI agents will need identity and trust on Nostr. Signet's credential system could verify that an AI agent is operated by a verified entity (Tier 3/4). This is a greenfield opportunity — nobody has shipped agent identity verification on Nostr yet.

The EU regulatory timeline is now a forcing function. EUDI Wallet mandate at end of 2026, interim age verification piloting in 5 countries, Animo shipping wallet infrastructure. Signet doesn't need to be the EUDI wallet — but it needs to be interoperable with VC standards coming out of W3C (Data Model v2.1, July 2026) and DIF. The Nostr DID Method is the bridge: if Signet credentials can be expressed as W3C VCs over Nostr DIDs, we participate in the EU ecosystem without abandoning decentralisation.

papiche's NIP-101 and fiatjaf's pyramid are the closest Nostr-native competitors. Both are building identity/trust layers on Nostr — NIP-101 with peer-validated credentials (overlaps Signet Tier 2 vouching) and pyramid with social proof ladders. Neither has Signet's professional verification tiers or ZKP layer, but they have active builders and could capture the "good enough" identity slot if Signet doesn't ship first.

SWOT

Strengths

Strength	Our repo(s)	Detail
Professional verification tiers	`signet`	No competitor has Tier 3/4 — professional body attestation is unique
ZKP age-range proofs (Bulletproofs)	`signet`	Cryptographic privacy, not ML/biometric — survives regulatory scrutiny
Deepfake-proof liveness (canary-kit)	`canary-kit`	CANARY-DERIVE word verification is novel — no biometric data stored
Nostr-native protocol	`signet`	No dependency on centralised wallets or DID registries
Guardian delegation model	`signet`	Kind 30477 with scoped delegation — ahead of competitors on child safety UX

Weaknesses

Weakness	Our repo(s)	Detail
Not yet published to npm	`signet`, `canary-kit`	Both use `file:` references — can't be consumed by external projects
No W3C VC interoperability	`signet`	Credentials are Nostr-native only — no bridge to VC Data Model v2.1
Single reference implementation	`signet`, `decented/decented`	Only Fathom — need more clients to prove protocol viability
No AI agent identity story	`signet`	NIP-XX and NIP-AC are shaping agent identity — Signet isn't in the conversation yet

Opportunities

Opportunity	Our repo(s)	Priority	Suggested action
AI agent identity verification on Nostr	`signet`	High	Draft a "Signet for AI Agents" design doc — how Tier 3/4 credentials verify agent operators
W3C VC interoperability via Nostr DID	`signet`	High	Investigate Nostr DID Method as bridge for Signet credentials → W3C VC ecosystem
NIP-XX EVM Identity Linking (#2262)	`signet`	Med	Comment on PR with Signet's Kind 30476 identity bridge design — show prior art
EU age verification pilot alignment	`signet`	Med	Review EU `av-doc-technical-specification` for compatibility with Signet's age-range proofs

Opportunity	Our repo(s)	Priority	Suggested action
OpenSats Nostr grant (Wave 16+)	`signet`	Med	Apply for Nostr grant — Signet is exactly the kind of protocol-level work they fund

Threats

Threat	Our repo(s) at risk	Severity	Detail
Google `longfellow-zk`	`signet`	High	FAANG distribution + Apache 2.0 — could become the default ZKP age verification stack
Animo `paradym-wallet`	`signet`	Med	Shipping EUDI wallet compliance fast — may capture the "verified identity on mobile" slot
papiche NIP-101	`signet`	Med	Nostr-native peer credentials — could become "good enough" identity if shipped first
fiatjaf `pyramid`	`signet`	Low	Social proof identity on Nostr — different approach but could reduce demand for formal verification
BetterAgeVerify	`signet`	Low	ML age verification — easier to integrate but weaker privacy guarantees

Action Items

Today

- **Review Google `longfellow-zk` source** — understand their ZKP approach, identify where Signet's Bulletproofs implementation differs or improves. Inform positioning. (`signet`)
- **Skim EU `av-doc-technical-specification`** — check if Signet's age-range proof output format could satisfy EU interim verification requirements. (`signet`)

This Week

- **Draft "Signet for AI Agents" design note** — AI agent identity on Nostr is wide open. Define how Signet credentials verify agent operators. Reference NIP-XX (#2226) and NIP-AC (#2253). (`signet` , design doc in `docs/plans/`)
- **Investigate Nostr DID Method interoperability** — can Signet credentials be expressed as W3C VCs over `did:nostr` ? This is the bridge to EU compliance without abandoning decentralisation. (`signet`)
- **Publish `canary-kit` to npm** — unblocks `signet-protocol` publication, which unblocks external adoption. Every week unpublished is a week competitors build distribution. (`canary-kit` , `signet`)
- **Comment on NIP-XX: On-Chain EVM Identity Linking (#2262)** — reference Signet Kind 30476 identity bridges as prior art. (`signet`)

Source Links

Item	Type	Link
google/longfellow-zk	Repo	https://github.com/google/longfellow-zk
Google ZKP age assurance blog	Blog	https://blog.google/technology/safety-security/opening-up-zero-knowledge-proof-technology-to-promote-privacy-in-age-assurance/
EU av-doc-technical-specification	Repo	https://github.com/eu-digital-identity-wallet/av-doc-technical-specification
EU av-web-verifier-ui	Repo	https://github.com/eu-digital-identity-wallet/av-web-verifier-ui
ageverification.dev	Web	https://ageverification.dev/
NIP-XX: AI Agent Messages	PR	https://github.com/nostr-protocol/nips/pull/2226
NIP-AC: DVM Agent Coordination v2	PR	https://github.com/nostr-protocol/nips/pull/2253
NIP-XX: On-Chain EVM Identity Linking	PR	https://github.com/nostr-protocol/nips/pull/2262

Item	Type	Link
NIP-XX: Dual Encryption / ECDH Delegated Access	PR	https://github.com/nostr-protocol/nips/pull/2258
NIP-24 shape field	PR	https://github.com/nostr-protocol/nips/pull/2268
NIP-43 membership state	PR	https://github.com/nostr-protocol/nips/pull/2267
W3C VC charter 2026	Spec	https://w3c.github.io/vc-charter-2026/
Nostr DID Method	Spec	https://nostrcg.github.io/did-nostr/
notedeck release	Release	https://github.com/damus-io/notedeck
didwebvh-rs	Repo	https://github.com/decentralized-identity/didwebvh-rs
trusted-ai-agents	Repo	https://github.com/decentralized-identity/trusted-ai-agents
trusted-ai-agents-wg	Repo	https://github.com/decentralized-identity/trusted-ai-agents-wg
labs-legally-binding-PoP-for-VCs	Repo	https://github.com/decentralized-identity/labs-legally-binding-proof-of-personhood-for-verifiable-credentials
pablof7z/ai-sdk-context-management	Repo	https://github.com/pablof7z/ai-sdk-context-management
tenex-chat/tenex	Repo	https://github.com/tenex-chat/tenex
pablof7z PR on vercel/ai	PR	https://github.com/vercel/ai/pulls?q=is%3Apr+author%3Apablof7z
fiatjaf/pyramid	Repo	https://github.com/fiatjaf/pyramid
papiche/NIP-101	Repo	https://github.com/papiche/NIP-101
papiche/UPassport	Repo	https://github.com/papiche/UPassport
animo/paradym-wallet	Repo	https://github.com/animo/paradym-wallet
animo/paradym-eudi-docs	Repo	https://github.com/animo/paradym-eudi-docs
animo/openid4vc-playground	Repo	https://github.com/animo/openid4vc-playground
BetterAgeVerify	Repo	https://github.com/BetterAgeVerify/BetterAgeVerify
AstroxNetwork/Proof-of-Personhood	Repo	https://github.com/AstroxNetwork/Proof-of-Personhood
nostrband/nostrlogin.org	Repo	https://github.com/nostrband/nostrlogin.org
privkeyio/keep	Repo	https://github.com/privkeyio/keep
Moonshots EP #236	Podcast	https://open.spotify.com/episode/2xLIHJDUHBzjKl55fERdCv
OpenSats Bitcoin grants (Mar 11)	Blog	https://opensats.org/blog
semaphore-protocol/semaphore-rs	Repo	https://github.com/semaphore-protocol/semaphore-rs

Sources: GitHub API (semaphore-protocol, decentralized-identity, animo, nostr-protocol orgs; decented, pablof7z, papiche, jb55, fiatjaf users), nostr-protocol/nips PRs, Ofcom Online Safety (403 — skipped), EU Digital Identity, PRIVO blog, OpenSats blog, W3C Credentials CG, Moonshots podcast, Stephan Livera Podcast, GitHub Trending (zero-knowledge, age-verification, self-sovereign-identity, verifiable-credentials, proof-of-personhood, nostr identity) Window: 2026-03-15 00:00 to 2026-03-16 17:00